Parameters

Server:

- Maximum Budget Q_s
- Period T_s

Task:

- Worst Case Execution Time C_I
- Minimum interarrival Time T_i

Variables

Server:

- Budget c_s
- Server deadline d_s
- Server Utilization/Bandwidth U_s = Q_s/T_s

Tasks:

- Hard Real-Time Task tau_i
- Job J_i
- Arrival time r_i
- Task deadline d_i
- f_i = finishing time of job

Rules

- Server is "active" if there exist served job J_i s.t. r_i <= t <= f_i
  Otherwise server is "idle"
- Served jobs are enqueued in a First in First Out queue

Pseudo Code Soft

```
New Job J_I Arrival:
    If [pending jobs]:
        enqueue new job
    Else if [U_s < c_s / (d_s – r_i)]:
        c_s  = Q_s
        d_s = r_i + T_s
    Start job


Budget exhausted:
    c_s = Q_s
    d_s = d_s + T_s
```

Definitions

Server:

- Budget c_s
- Maximum Budget Q_s
- Period T_s
- Server deadline d_sk

Tasks:

- Hard Real-Time Task tau_i
- Job J_ij
- Arrival time r_ij
- Task deadline d_ij
- Ci = WCET
- Ti = minimum interarrival time

Pseudo Code Soft

```
New Job Arrival:
   If [pending jobs]:
      enqueue new job
   Else if [U_s < c_s / (d_s – r)]:
      c_s  = Q_s
      d_s = r + T_s
   Start job


Budget exhausted:
   c_s = Q_s (replenish)
   d_s = d_s + T_s (reset deadline)
```

Example
Task 1: ^oo^oo
Server: 111111

Task Parameter Table

Tasks C T

Tau_1 1 3

Server Parameter Table

Q_s 1

T_s 3

Server Variable Table

Budget ?

Server Deadline ?

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

Else if [U_s < c_s / (d_s – r)]:

c_s = Q_s

d_s = r + T_s

> Start job

Budget exhausted:

c_s = Q_s (replenish)

d_s = d_s + T_s (reset deadline)

Example

Task 1: ^xo^xo
Task 2: ^ox^ox
Server: 321321

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	3
T_s	3

Server Variable Table

Budget ?

Server Deadline ?

Utilization ?

Pseudo Code Soft

New Job J_i Arrival:

    If [pending jobs]:
        enqueue new job
    > Else if [U_s < c_s / (d_s – r_i)]:
        c_s = Q_s
        d_s = r + T_s
    Start job

Budget exhausted:

    c_s = Q_s
    d_s = d_s + T_s

Example

Task 1: ^oo^oo
Task 2: ^xo^xo
Server: 111111

Task Parameter Table

Tasks	C	T
Tau_1	1	3
Tau_2	1	3

Server Parameter Table

Q_s	1
T_s	3

Server Variable Table

Budget ?

Server Deadline ?

Utilization ?

Pseudo Code Soft

New Job Arrival:

If [pending jobs]:

enqueue new job

> Else if [U_s < c_s / (d_s – r)]:

c_s = Q_s

d_s = r + T_s

Start job

Budget exhausted:

c_s = Q_s

d_s = d_s + T_s